

IS390 Progress Report: CLEAR Streaming CL + iNeMo-Inspired Proxy

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Abstract

This stage progress report summarizes the current implementation and preliminary evaluation of an IS390 continual-learning project built around a CLEAR-style streaming protocol. The project develops a compact PyTorch benchmark in which a model trains on temporal bucket i and evaluates on bucket $i+1$, thereby approximating next-domain evaluation under distribution shift. The current system supports synthetic CLEAR-style image generation, bucket discovery, configurable ResNet-18 training, replay memory, prototype memory, prototype-alignment regularization, latent-space separation, per-run artifact generation, and suite-level aggregation across four ablation settings. The current experiments use a locally generated synthetic proxy dataset rather than the official CLEAR dataset, so the results should be interpreted as pipeline-validation evidence rather than final scientific findings. In the May 8 delivery suite, `replay_plus_prototypes` achieved the highest mean next-domain accuracy at 0.425, closely followed by `full_inemo_proxy` at 0.424 and `finetune_only` at 0.415, while `replay_baseline` reached 0.357. Forgetting remained at 0.0 across the suite, suggesting that the current synthetic setting may be too simple or too separable to expose strong retention challenges. The next stage should prioritize real CLEAR-10 integration, multi-seed evaluation, variance reporting, stratified shadow holdout, and automated testing infrastructure.

Research Motivation and Problem Statement

Continual learning addresses the problem of adapting a model to sequentially arriving data without losing performance on previously encountered distributions. A central challenge is catastrophic forgetting: after training on newer data, a model may degrade on older domains or classes. This project focuses on a CLEAR-style streaming scenario in which data is organized into temporal or domain-shifted buckets and evaluation emphasizes the model's ability to generalize to the next bucket in sequence.

The research goal at this stage is not to claim a new state-of-the-art continual-learning method. Instead, the goal is to build a runnable, inspectable, and reproducible research infrastructure that can compare fine-tuning, replay, prototype-based regularization, and combined methods under a consistent next-domain evaluation protocol. The current implementation therefore functions as a platform for controlled ablation studies before moving to the official CLEAR dataset and broader experimental validation.

Stage Objectives

The current stage of the project is focused on implementation readiness and preliminary experiment validation. The stage objectives are:

1. Implement a runnable CLEAR-style streaming pipeline that preserves the documented command-line entry point.
2. Support synthetic proxy data while maintaining compatibility with real CLEAR-style bucketed datasets.
3. Compare four ablation conditions: fine-tuning only, replay baseline, replay plus prototypes, and a full iNeMo-inspired proxy.
4. Generate reproducible experiment artifacts, including per-run summaries, CSV metrics, plots, and suite-level aggregation outputs.
5. Identify research and engineering limitations before moving to official CLEAR data and multi-seed evaluation.

Research System Architecture and Methodology

The implementation follows a modular research-system design. The `src/run.py` entry point loads the YAML configuration, seeds the runtime, generates or discovers data buckets, constructs loaders, builds the model, instantiates the streaming trainer, executes training and evaluation, and emits structured artifacts. The surrounding modules separate data handling, model definition, memory mechanisms, loss functions, metrics, and reporting logic.

System Architecture

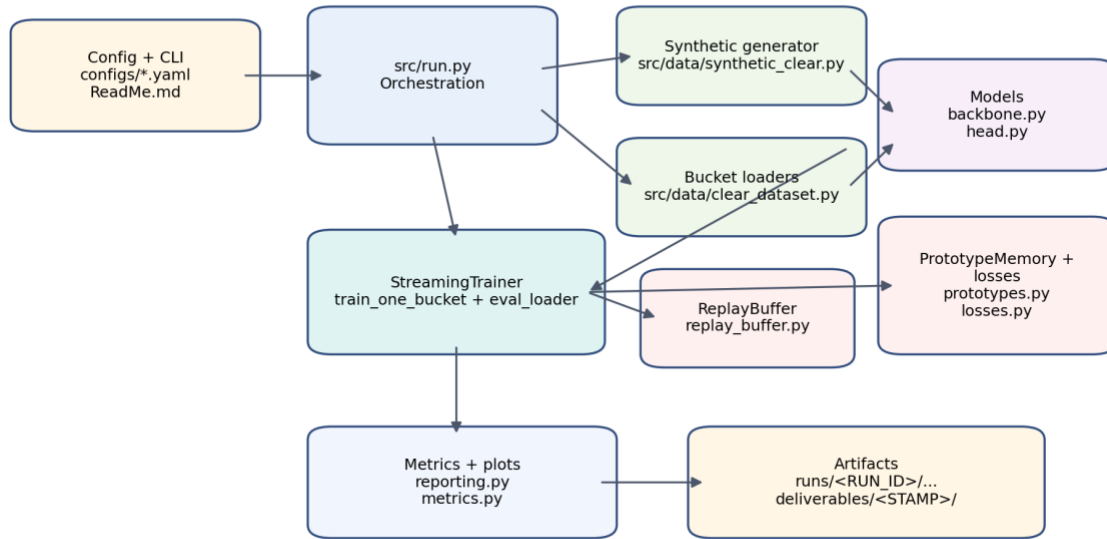


Figure 1. Research system architecture derived from the repository source files.

The execution workflow begins with installation and configuration, then proceeds through dataset loading, bucket-wise model training, next-bucket evaluation, artifact writing, and suite-level aggregation. This staged workflow is appropriate for a research project because each step creates inspectable outputs that can be validated independently.

Execution Workflow



Figure 2. Execution workflow from configuration to research artifacts.

Dataset and Experimental Setting

The current experiments use synthetic generated images, not the official CLEAR dataset. The default configuration points to `data/synth\_CLEAR\_delivery\_v1`, a locally generated CLEAR-style proxy dataset with bucket-wise domain shift. The synthetic proxy is useful for validating the research pipeline, but it does not support claims about performance on real-world CLEAR imagery.

Property	Current Setting	Research Interpretation
Dataset root	<code>./data/synth_CLEAR_delivery_v1</code>	Local proxy dataset used for controlled pipeline validation.
Number of buckets	6	Sequential domains for streaming training and next-bucket evaluation.
Number of classes	11	Multi-class visual recognition setting.
Images per class per bucket	40	Balanced synthetic sampling within each

		bucket.
Total images	2,640	$6 \times 11 \times 40$ generated images.
Image size	64×64	Small images enabling fast iteration.
Layout	imagefolder	Bucket/class directory structure compatible with common image loaders.

Table 1. Current synthetic CLEAR-style proxy dataset used in the stage experiments.

The repository also contains smaller synthetic datasets and manifest-layout examples. These assets support loader validation and controlled experiments, while the default delivery benchmark remains `synth\_CLEAR\_delivery\_v1`.

Dataset Path	Buckets	Layout	Images	Image Size
data/synth_CLEAR	6	imagefolder	1,320	224×224
data/synth_CLEAR_delivery_v1	6	imagefolder	2,640	64×64
data/synth_CLEAR_manifest	3	manifest_csv	60	32×32
data/synth_CLEAR_stage1	3	imagefolder	60	32×32

Table 2. Local synthetic datasets discovered in the repository inventory.

Data and File Flow

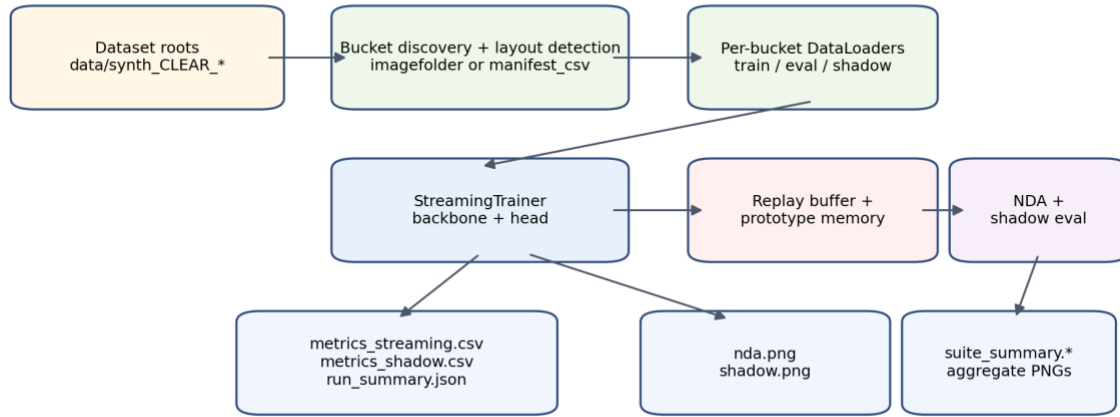


Figure 3. Data and file flow across dataset inputs, trainer state, and generated outputs.

Model and Training Configuration

The current model configuration uses a ResNet-18 backbone with a 512-dimensional representation and a linear classification head. The number of output classes is inferred automatically from the dataset. Training is configured for fast iteration rather than final performance maximization: each bucket is trained for two epochs with a batch size of 64, learning rate 0.001, and weight decay 0.0005. CUDA is requested in the configuration, so portability would improve if a documented CPU or auto-device fallback were added.

Configuration Area	Current Value	Purpose
Backbone	ResNet-18	Compact convolutional model for visual classification.
Feature dimension	512	Latent representation used by the linear head and prototype memory.
Classification head	Linear	Simple head supporting interpretable ablations.
Device	CUDA	GPU acceleration for training.
Batch size	64	Moderate training batch size for small images.
Epochs per bucket	2	Fast bucket-wise iteration.
Learning rate	0.001	Baseline optimizer step size.
Weight decay	0.0005	Regularization for model weights.

Table 3. Model and training settings used by the current stage configuration.

Continual Learning Methods and Ablations

The project compares four experimental conditions that isolate the effect of replay and prototype-based regularization. The terminology in this report uses “iNeMo-inspired proxy” because the repository implements prototype memory, prototype alignment, and latent separation mechanisms inspired by prototype-based continual learning rather than claiming a complete reproduction of the original iNeMo method.

Ablation	Replay	Prototype Regularization	Latent Separation	Research Role
finetune_only	No	No	No	Lower-complexity baseline without memory mechanisms.
replay_baseline	Yes	No	No	Tests replay memory alone.
replay_plus_prototypes	Yes	Yes	No/partial	Tests replay combined with class-prototype guidance.
full_inemo_proxy	Yes	Yes	Yes	Tests the full implemented iNeMo-inspired proxy.

Table 4. Research ablation structure used to compare continual-learning mechanisms.

The replay configuration uses reservoir sampling with a buffer size of 2,000, replay ratio of 0.30, alpha value of 1.0, and per-class cap of 50. This is intended to preserve examples from earlier buckets and reduce forgetting during later bucket training.

The prototype configuration maintains class prototypes with momentum 0.9 and prototype alignment strength 0.05. The latent partition component adds a margin-based separation term with strength 0.03 and margin 0.15. Together, these mechanisms encourage features to remain close to their class prototypes while preserving separation between classes in the latent space.

Evaluation Protocol and Metrics

The evaluation protocol follows a CLEAR-style streaming design: the model trains on bucket i and evaluates on bucket $i+1$. The primary metric is next-domain accuracy (NDA), which measures performance on the immediately following domain after each training step. This metric is appropriate for assessing short-horizon generalization under temporal or domain shift.

The configuration also enables a shadow holdout ratio of 0.20 and forgetting computation. During execution, buckets are converted into training, full-evaluation, and optional shadow loaders. The trainer writes streaming metrics, optional shadow metrics, run summaries, class maps, and plots into timestamped run directories. Suite-level scripts aggregate the four ablations into summary tables and cross-ablation figures.

Artifact	Purpose
config_resolved.yaml	Records the exact resolved configuration for a run.
metrics_streaming.csv	Stores NDA and related streaming metrics by step.
metrics_shadow.csv	Stores optional shadow holdout accuracy and forgetting metrics.
run_summary.json	Summarizes run-level metrics and metadata.
class_map.json	Maps class labels to numeric IDs when enabled.
nda.png	Per-run next-domain accuracy plot.
shadow.png	Per-run shadow holdout plot when available.
suite_summary.csv	Machine-readable cross-ablation summary.
aggregate_plots	Suite-level figures for comparing ablation behavior.

Table 5. Metrics and artifacts generated by the current evaluation pipeline.

Preliminary Results

Benchmark visuals and summary values are drawn from the May 8, 2026 delivery suite. Across the four ablations, replay_plus_prototypes achieved the highest mean next-domain accuracy at 0.425, narrowly followed by full_inemo_proxy at 0.424 and finetune_only at 0.415. The replay_baseline condition achieved 0.357. Because these are single-seed results on synthetic data, the top-three ordering should be treated as a preliminary stage finding rather than a statistically supported conclusion.

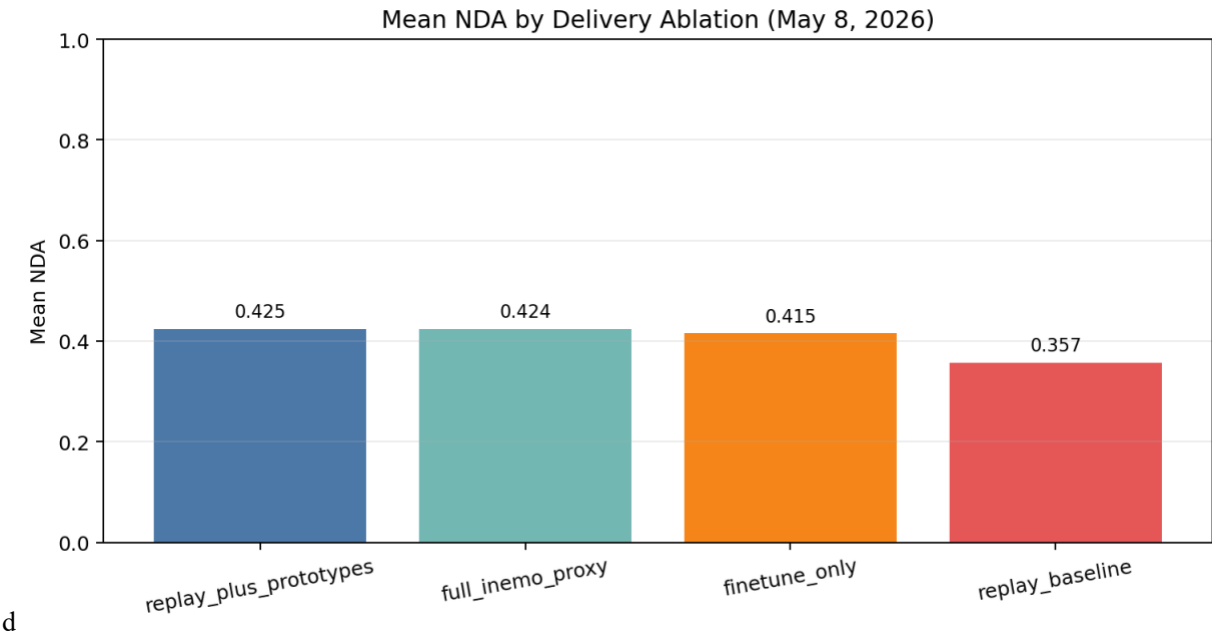


Figure 4. Mean next-domain accuracy by ablation from the May 8, 2026 delivery suite.

The step-wise trajectory shows comparable behavior across several configurations in earlier steps, with replay_plus_prototypes and full_inemo_proxy pulling ahead at later steps. The replay_baseline curve regresses at step 4, which helps explain its lower mean NDA. Forgetting remained at 0.0 across the suite, likely reflecting the simplicity or separability of the current synthetic proxy rather than demonstrating robust forgetting resistance.

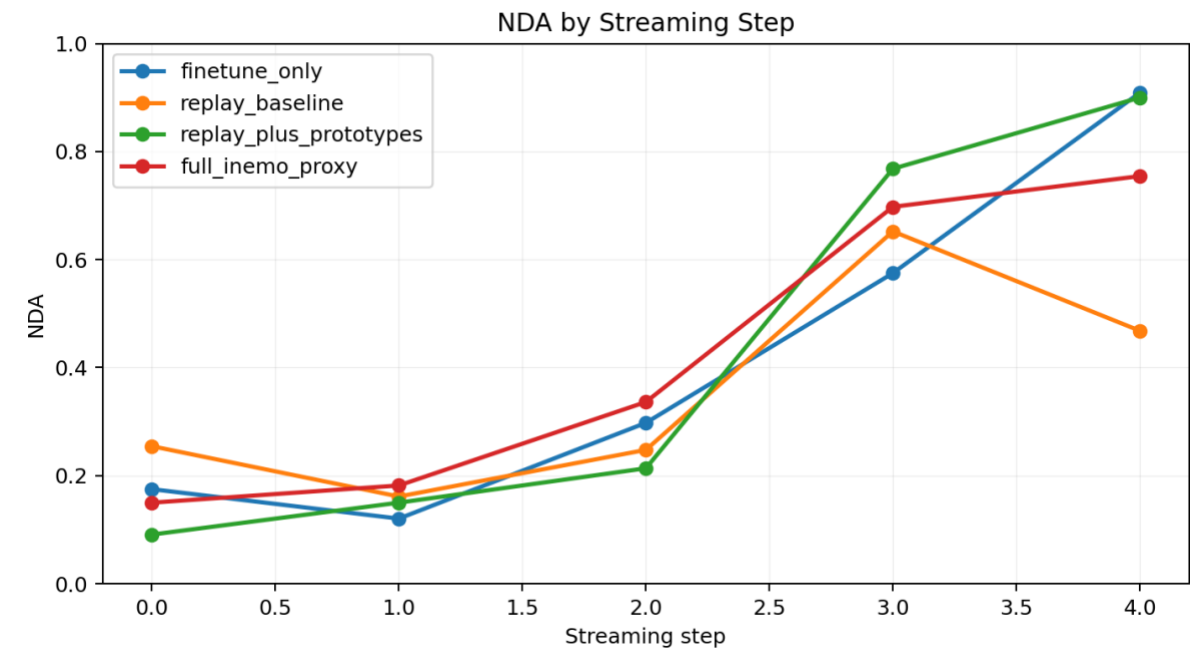


Figure 5. NDA by streaming step across the four reported ablations.

Discussion

Current evidence suggests that combining replay with prototype guidance is promising in this controlled benchmark, but the margin between `replay_plus_prototypes` and `full_inemo_proxy` is too small to support a strong method-level conclusion. The similarity between those two settings may indicate that prototype alignment captures most of the benefit in the current synthetic setup, while the additional latent-separation term provides limited incremental value under these conditions.

The relatively weaker `replay_baseline` result may indicate that replay alone is not sufficient under the chosen sampling, training length, and proxy distribution. However, this interpretation remains tentative because the benchmark is synthetic and single-seed. The absence of measurable forgetting is especially important: it suggests that the current dataset may not impose enough long-term retention pressure, or that the task is separable enough for all methods to retain prior knowledge under the current settings.

The results therefore validate the experiment pipeline more strongly than they validate the scientific superiority of any particular method. The next research stage should increase external validity through official CLEAR-10 integration and improve statistical validity through multi-seed evaluation.

Current Limitations

The current limitations should be interpreted as stage-progress constraints rather than failures. They identify what must be strengthened before the project can support stronger scientific conclusions.

Limitation	Type	Implication
Synthetic dataset only	Research validity	Results should not be generalized to real CLEAR performance.
No official CLEAR integration yet	Research validity	External validity remains incomplete.
Single-seed results	Statistical validity	No confidence intervals or variance estimates are available.
Forgetting remains 0.0	Benchmark difficulty	The proxy may be too simple to stress retention mechanisms.
No automated tests or CI	Engineering reliability	Regression detection relies on manual execution.
No lockfile or containerization	Reproducibility	Exact environment reproduction is not guaranteed.
Random shadow split	Evaluation design	Class coverage may be uneven in small buckets.
Stdout parsing in suite runner	Workflow robustness	Console-message changes could break aggregation.

Table 6. Current research and engineering limitations of the stage implementation.

Next Stage Plan

The next stage of this project should focus on strengthening both the scientific validity and engineering reliability of the benchmark. First, the synthetic CLEAR-style proxy dataset should be extended or replaced with the official CLEAR-10 dataset so that results better reflect real temporal domain shift. Second, the ablation suite should be rerun across multiple random seeds, with variance or confidence intervals reported alongside mean NDA. The evaluation pipeline should also be improved by adding class-stratified shadow holdout splits, clearer forgetting analysis, and additional continual-learning baselines. On the engineering side, adding smoke tests, CI checks, and a more stable artifact contract for suite execution would make the repository easier to reproduce, maintain, and extend in future research stages.

Conclusion

This stage of the IS390 project has produced a runnable CLEAR-style continual-learning benchmark with modular source structure, synthetic data generation, configurable ablations, next-domain streaming evaluation, per-run artifacts, and suite-level reporting. The implementation is sufficient for demonstration, controlled ablation comparison, and research-progress documentation.

The current scientific findings should remain preliminary. The strongest current evidence is that the pipeline runs end-to-end and that replay plus prototype-based regularization can be compared under a consistent protocol. The next scientific priority is to validate the system on real CLEAR-style data, repeat the suite across multiple seeds, add variance and confidence reporting, and strengthen the engineering infrastructure with tests, CI, and reproducible environment specifications. These steps would move the project from a validated stage prototype toward a more reliable research benchmark.